

Siyuan Lu

✉ siyuan.lu • ✉ lusiyanzs@gmail.com • 🐱 IcyFish332

🎓 Education

Zhejiang University / *Computer Science and Technology* / *Ph.D.* 2025 – 2030 (Expected)

Joint program with Westlake University and Shanghai Innovation Institute

Research: Agentic Systems and Agent Training Advisors: *Tao Lin, Weinan Zhang*

Beijing University of Posts and Telecommunications / *Internet of Things* / *B.E.* 2021 – 2025

Joint program with Queen Mary University of London

Weighted Average Mark: 90.97/100 Rank: 7/182

📖 Publications

Don't Just Fine-tune the Agent, Tune the Environment

First Author ICLR 2026

Proposed Environment Tuning paradigm that enables agents to learn complex behaviors directly from problem instances without pre-collected expert trajectories. Through structured curriculum, environment augmentation, and fine-grained rewards, achieved superior tool-calling capabilities with only 400 samples.

AWorld: Orchestrating the Training Recipe for Agentic AI

Co-author Technical Report

Proposed an open-source system for large-scale agent-environment interaction, achieving 14.6x speedup in experience collection through distributed deployment. The trained Qwen3-32B model reached 32.23% accuracy on GAIA test set, surpassing GPT-4o.

Unleashing the Potential of Large Language Models as Prompt Optimizers: An Analogical Analysis with Gradient-based Model Optimizers

Fourth Author AACL 2025

Proposed GPO (Gradient-inspired Prompt Optimizer) by analogizing gradient-based model optimizers to design LLM prompt optimizers. Achieved up to 56.8% and 62.6% performance improvements on Big-Bench Hard and MMLU respectively.

PreAct: Predicting Future in ReAct Enhances Agent's Planning Ability

Third Author COLING 2025

Proposed PreAct framework that integrates prediction, reasoning, and action. By leveraging predictive information, enables agents to provide broader and more strategic reasoning, outperforming ReAct methods on complex tasks.

🏢 Work Experience

Ant Group

2025.6 – Present

Research Intern AWorld Team, InclusionAI

Mentor: *Chenyi Zhuang*

Participated in research and development of the AWorld project, producing an ICLR conference paper on post-training of large models in the tool-calling domain.

👥 Projects

AWorld

2025.6 – Present

Contributor Agentic AI, Self-Evolution, Multi-Agent Systems

AWorld is a self-evolving agent framework providing a complete agent learning loop (environment access, agent construction, experience retrieval, model training). The framework supports multi-agent collaboration, rich tool integration paradigms, advanced context management, and event-driven architecture, capable of capturing complete execution trajectories for continuous optimization. Main contributions: Participated in designing the training module, enabling training frameworks (such as verl) to leverage AWorld's agent abstraction and distributed environment deployment capabilities, accelerating rollout speed for large model reinforcement learning training and improving training efficiency.

MASArena

2025.2 – 2025.8

Core Maintainer & Contributor Multi-Agent Systems, Benchmarking, LLM

MASArena is a comprehensive single/multi-agent system benchmarking framework supporting agent performance evaluation across multiple tasks (math, code, reasoning, etc.). The framework adopts modular design with built-in automated workflow optimization system, using LLM-driven evolutionary algorithms to optimize agent workflows, and provides visualization debugging and failure attribution analysis capabilities.

Honors

First-Class Scholarship, Beijing University of Posts and Telecommunications 2023

Second-Class Scholarship, Beijing University of Posts and Telecommunications 2022

Skills

IELTS 7.0, proficient in English listening, speaking, reading, and writing

Familiar with Linux systems, proficient in common Linux commands, daily development in Linux environment

Familiar with Git, Docker, proficient in using GitHub for version control

Familiar with modern Python development, proficient in using latest Python toolchain such as uv, ruff

Familiar with PyTorch, HuggingFace Transformers, DeepSpeed, vllm and other deep learning frameworks